

SARTHAK JAYANT WALKE

Mechanical Engineering Student | AI/ML for Mechanical Systems

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PROFESSIONAL SUMMARY

Third-year Mechanical Engineering student with a strong foundation in thermodynamics, fluid mechanics, and IC engines, combined with hands-on skills in Python, machine learning, and automation tool development. Passionate about the intersection of AI and mechanical engineering — particularly predictive maintenance and robotics — with a goal of using AI/ML to solve real-world mechanical and industrial problems.

EDUCATION

A.I.S.S.M.S. College of Engineering, Pune

2024 – 2028

Bachelor of Engineering – Mechanical Engineering | CGPA: 8.3 (1st Yr), 8.0 (2nd Yr)

TECHNICAL SKILLS

Languages: Python, C, SQL

AI/ML: Machine Learning fundamentals, Data Analysis, Model Building

Engineering & Design: CAD Modeling & Assembly, Applied Thermodynamics, Fluid Mechanics, IC Engines

Tools: Google Colab, Jupyter Notebook, Git, GitHub, VS Code, MATLAB, Excel/Tally-compatible reporting, PDF data extraction

AREAS OF INTEREST

AI/ML for Mechanical Systems • Predictive Maintenance • Robotics & Automation • Intelligent Manufacturing • Machine Learning for Fault Detection

PROJECTS

Design and Structural-Thermal Performance Analysis of an Automotive Wheel Rim Using A356-T6, Al 6061-T6 and Al 7075-T6 (Research Paper)

- Authored a research paper evaluating structural and thermal performance of automotive wheel rims across three aluminum alloys — A356-T6, Al 6061-T6, and Al 7075-T6.
- Modeled the wheel rim and carried out comparative structural-thermal simulation to assess strength, thermal response, and material suitability.
- Analyzed and compared results across alloys to identify performance trade-offs relevant to automotive wheel design.

AI-Assisted Airfoil Design and Aerodynamic Optimization Using CAD, CFD, MATLAB and Machine Learning

- Ongoing project combining CAD modeling, CFD simulation, and machine learning to optimize airfoil aerodynamic performance.
- Using CAD for airfoil geometry generation and CFD tools for aerodynamic flow simulation and analysis.
- Applying MATLAB for data processing and analysis, paired with machine learning models to assist in aerodynamic optimization.

ML-Based Predictive Maintenance of Rotating Machinery

- Analyzed vibration/sensor data to identify operating conditions and potential machine faults.
- Performed data preprocessing, feature extraction, and exploratory analysis using Python.
- Developed and evaluated ML models for fault classification using Scikit-learn.

Predicting Engine Fuel Efficiency using Machine Learning

- Analyzed the relationship between engine parameters and fuel efficiency using a dataset of cylinders, horsepower, weight, and acceleration.
- Built and evaluated a machine learning model to accurately predict engine performance from these input variables.

EEG-Based Smart Helmet Safety System – Highway Hypnosis Detection

Team Project, Electronics Instruments & Measuring Systems Course

- Designed a smart helmet system using an EEG sensor to monitor rider focus/drowsiness in real time and prevent highway-hypnosis-related accidents.

- Implemented button-free gesture-based ignition control, personal focus-baseline calibration, and a two-level safety intervention (buzzer/LED warning, then automatic ignition cutoff).
- Demonstrated retrofit capability by integrating and testing the system on an old scooter, independent of vehicle model or manufacturing year.

Full CAD Assembly of a Robot

- Modeled and assembled a complete robot in CAD, including individual part design and full assembly integration.

Data-Shift – Automated Accounting Tool for CA Firms

- Built a Python tool that extracts key data (GST details, dates, party names, tax, amounts) from uploaded PDF invoices and bank statements and converts it into structured, Tally-import-compatible Excel sheets.
- Implemented secure user login and automated the data processing pipeline, significantly reducing manual accounting work for the client.

LEADERSHIP & EXTRACURRICULAR

Media Head, GDG (Google Developer Groups) – AISSMS COE Chapter

INTERNSHIP AVAILABILITY

Current Year/Semester: 3rd Year, 5th Semester

Tentative Duration: 8–10 weeks (flexible, subject to mutual availability)

CERTIFICATIONS

CAD Design Course – Udemy

MATLAB Onramp – MathWorks